



Data Visualization

Milestone 2

COM-480

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1 | Project Goal

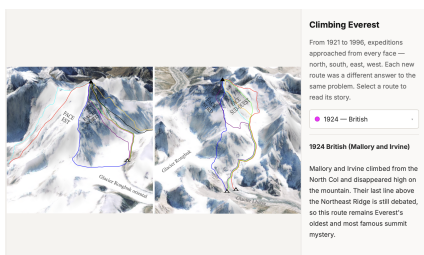
We trace how Himalayan climbing went from rare national expeditions to a commercialized industry. The Himalayan Database (468 peaks, 10,364 expeditions, 76,519 climbers, 1,106 deaths) lets us show the growth in expeditions, the rise of trekking agencies, and a death rate that has not dropped much, hitting hired staff hardest, most of whom are Nepali.

2 | Outline

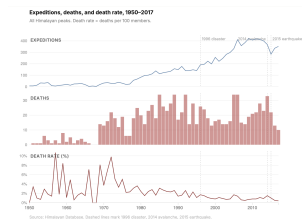
We ask three questions: how has participation in Himalayan climbing changed, how have the risks shifted, and when do people die? The visualizations explore routes, temporal trends, and the distribution of deaths.

Figure 2.1a is the route visualization. Routes sit on an SVG map of Everest next to a panel with expedition details. A dropdown sorted by year lists 16 historical routes (1924–1996) with color swatches and notes. This version is static. For the next milestone we will replace it with an interactive 3D version where users can click routes and see infographics.

Figure 2.1b is a three-panel small-multiples chart (shared x-axis, 1950–2017) showing expeditions, deaths, and death rates. Expeditions grew fast, but the death rate stayed flat. Dashed annotations mark the 1996 disaster, 2014 avalanche, and 2015 earthquake. We plan to add panels for agency share and success rate to build out the commercialization argument.



(a) Sketch of Everest route visualization.



(b) Expeditions, deaths, and death rate over time.

Figure 2.1: Overview of route visualization and expedition trends.

Figure 2.2 maps deaths by altitude. Each dot is one fatality placed at the altitude where it happened, split between hired staff and clients. Four scroll-driven steps build up the picture: all Himalayan deaths in gray ($n=957$), then colored by role with median altitude lines, then filtered to Everest ($n=299$, median gap of $\sim 2,400$ m), and finally a death-rate callout (hired 2.0%, clients 1.3%). Hovering over a dot shows cause, altitude, year, peak, citizenship, and hired status.

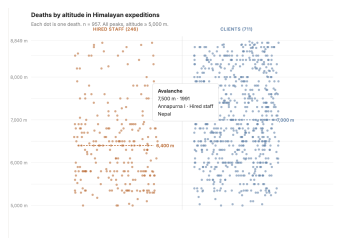


Figure 2.2: Distribution of deaths by altitude in Himalayan expeditions.

We also plan a visualization that tracks how the climber demographic changed with commercialization. We have 73,022 members with age (median 36, range 7–85), gender (90.8% male), and nationality. The chart would frame these shifts around the growth of agencies: how the age distribution and gender ratio moved

as commercial operators took over. Nationality filters would let users compare how different climbing populations changed over time.

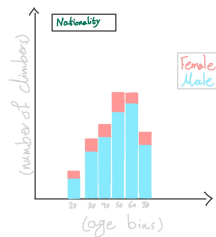


Figure 2.3: Demographic shift over decades.

3 | Technologies

No front-end framework. Plain HTML, CSS, and vanilla JavaScript with **D3.js v7** (CDN) for charts and **Scrollama** (CDN) for scroll-driven navigation. The Chapter 2 map is a public-domain SVG from Wikimedia Commons. Data preprocessing is Python with **pandas**. Typography: **Google Fonts (Inter)**. Hosting: **GitHub Pages** (static files from docs/ on the main branch). For Milestone 3, Chapter 2 will use **Three.js** for 3D terrain.

Lectures used:

1. Web Development (L1–2) — site structure and layout.
2. JavaScript (L2–3) — data loading and DOM manipulation.
3. D3.js (L4) — scales, axes, and layouts for the charts in Chapters 3 and 4.
4. Interaction (L5) — scroll-driven navigation, tooltips, dropdown controls.
5. Perception and Color (L6) — color encoding for hired/client roles, altitude mapping.

4 | Future Work

For Milestone 3 we plan to implement:

- **Interactive 3D Everest.** Replace the flat SVG with a rotatable Three.js terrain where routes follow real elevation and open infographics on click (first ascent, team, fatalities). This is the centrepiece.
- **Narrative writing.** Complete the scroll text for every step: the opening, the 1953 and 1996 turning points, and the closing argument about the cost of commercialization.
- **Ch.3 expansion.** Add agency-share and success-rate panels to strengthen the commercialization argument.
- **Visual polish.** Responsive/mobile layout, improved Chapter 1 visuals, smoother chapter transitions.

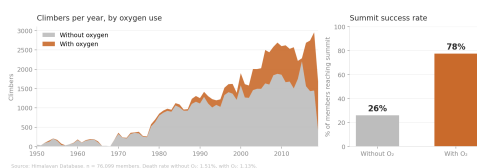
5 | Additional Ideas

If time allows, we will explore several improvements of our core visualizations:

1. **Mountain-silhouette background.** A faint Everest silhouette behind the dots in the Cost section, aligned to the altitude axis, implicitly labelling the death zone above 8 000 m.

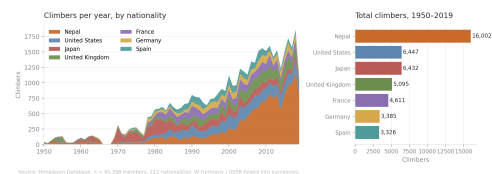
2. **Death-cause layer.** Colour the dots in the Cost section by **death_cause** (avalanche, fall, altitude sickness, illness).
3. **Oxygen dependence timeline.** Integration of oxygen use data and the gap between teams with and without oxygen and its effect on climbing and deaths.
4. **Animated route tracing.** Draw routes progressively in chronological order from 1921, using a D3 transition on each route keyed on the **year** field.
5. **Nationality origin map.** Show the shift from a handful of Western expeditions to a broader international client base, with Nepali hired staff forming the dominant base layer.

Oxygen dependence in Himalayan climbing, 1950-2019



(a) Oxygen dependency data prototype.

Where climbers come from, 1950-2019



(b) Nationality dependency data prototype.

Figure 5.1: Prototypes exploring oxygen use and nationality effects.